

ON FANO MANIFOLDS WITH THE INTEGRAL COHOMOLOGY RING OF PROJECTIVE SPACE

ABSTRACT. Fujita conjectured that a complex Fano manifold with the integral cohomology ring of complex projective space is biholomorphic to projective space. We prove the conjecture in dimensions at most ten. We also prove it in arbitrary dimension n when the Fano index is at least $n - 3$, and in even dimension when the index is at least $n - 7$. The proofs combine Hirzebruch–Riemann–Roch with bounds on spaces of sections. These give finitely many possible Hilbert polynomials, which are excluded by characteristic-number identities and congruences. Two cases require an additional argument using section rings of genus-two curves. The finite calculations are carried out by an accompanying Python program.

1. INTRODUCTION

A compact Kähler manifold homeomorphic to $\mathbb{P}^n$ is biholomorphic to $\mathbb{P}^n$. This characterization follows from work of Hirzebruch and Kodaira, Novikov, and Yau; see [Deb15, Theorem 1]. Its proof uses the rational Pontryagin classes, which are determined by the homeomorphism type, in a Riemann–Roch calculation. An isomorphism of integral cohomology rings does not prescribe these classes.

Recall that a *Fano manifold* is a smooth projective complex manifold whose anticanonical line bundle K_X^{-1} is ample. Fujita formulated the following conjecture in [Fuj80, §1].

Conjecture 1.1 (Fujita, C_n). *Let X be a Fano manifold of dimension n with*

$$H^*(X; \mathbb{Z}) \simeq H^*(\mathbb{P}^n; \mathbb{Z})$$

as graded rings. Then X is biholomorphic to $\mathbb{P}^n$.

The ring structure is essential. An odd-dimensional smooth quadric has the same integral cohomology groups as projective space, but in dimension at least three the top power of its ample generator has degree two. For projective space this degree is one, and it is this restriction that will give bounds on spaces of sections in our argument.

Fujita proved C_n for $n \leq 5$ [Fuj80, Theorem 1]. The work of Libgober and Wood [LW90] extends the relevant characteristic-number arguments to dimension six. Debarre [Deb15, Theorem 2] gives the resulting classification under the integral cohomology-ring hypothesis: in dimensions at most six, a compact Kähler manifold with this ring is projective space, with the possible exception of ball quotients in dimensions four and six. The latter have ample canonical bundle and hence do not occur in the Fano case. In dimension seven, Debarre’s calculation leaves the possibilities $c_1(X)^7 \in \{\pm 2^7, \pm 4^7\}$ besides projective space [Deb15, Theorem 10]. For a Fano manifold these are the two positive indices 2 and 4.

Fujita’s conjecture arose together with his conjectures A_n and B_n on compactifications and hyperplane sections. More recently, Li and Peternell [LP25] proved that a compact Kähler manifold containing a smooth connected divisor with homologically trivial complement is projective space with a hyperplane, provided $n \not\equiv 3 \pmod{4}$. Their hypothesis includes a distinguished divisor. Conjecture C_n assumes only the cohomology ring and positivity of the anticanonical bundle, so such a divisor is not available at the outset.

We prove the following case of Fujita’s conjecture.

Theorem 1.2. *Let X be a Fano manifold of dimension $n \leq 10$ with $H^*(X; \mathbb{Z}) \simeq H^*(\mathbb{P}^n; \mathbb{Z})$ as graded rings. Then X is biholomorphic to $\mathbb{P}^n$.*

The *Fano index* of a Fano manifold X is the largest positive integer r for which

$$K_X^{-1} \simeq A^{\otimes r}$$

for some holomorphic line bundle A . Such a line bundle A is necessarily ample. Under the cohomology-ring hypothesis of Conjecture 1.1, the first Chern class identifies $\text{Pic}(X)$ with $H^2(X; \mathbb{Z}) \simeq \mathbb{Z}$. If L is its ample generator and $x = c_1(L)$, then the index is the positive integer r determined by

$$K_X^{-1} \simeq L^{\otimes r}, \quad c_1(T_X) = rx, \quad \int_X x^n = L^n = 1.$$

We also obtain the following result in arbitrary dimension.

Theorem 1.3. *Let X be a Fano manifold of dimension n with $H^*(X; \mathbb{Z}) \simeq H^*(\mathbb{P}^n; \mathbb{Z})$ as graded rings, and let r be its Fano index. Then X is biholomorphic to $\mathbb{P}^n$ if either of the following conditions holds:*

- (i) $r \geq n - 3$;
- (ii) n is even and $r \geq n - 7$.

1.1. Outline of the proof. By the theorem of Kobayashi and Ochiai [KO73], a Fano n -fold has index at most $n + 1$, with equality precisely for $\mathbb{P}^n$. To prove Theorems 1.2 and 1.3, it therefore suffices to exclude the possible indices $r < n + 1$ under their respective hypotheses.

We first use the cohomology ring to restrict r . The Hodge numbers are those of $\mathbb{P}^n$, so $\chi(X, \Omega_X^p) = (-1)^p$. Writing $c_j(T_X) = c_j x^j$, Riemann–Roch and integrality give

$$c_n = n + 1, \quad rc_{n-1} = \frac{n(n+1)^2}{2}, \quad r \equiv n + 1 \pmod{2}. \quad (1.1)$$

In particular, r is a positive divisor of $n(n+1)^2/2$, of the specified parity, and at most $n + 1$. In dimensions seven through ten the possibilities are as follows.

n	indices to exclude	index of projective space
7	2, 4	8
8	1, 3	9
9	2, 6	10
10	1, 5	11

For each remaining index we consider the Hilbert polynomial $P(t) = \chi(X, L^t)$. Kodaira vanishing gives

$$P(-1) = \cdots = P(-(r-1)) = 0, \quad P(j) = h^0(X, L^j) \quad (j \geq 0),$$

and Serre duality gives $P(-r-t) = (-1)^n P(t)$. These identities, the leading coefficient $1/n!$, and $P(0) = 1$ determine P from the first few section dimensions $h^0(X, L^j)$; the precise parametrization is given in Section 5. The intersection-theoretic bound

$$h^0(X, L^j) \leq j^n + n$$

then puts these parameters in finite intervals. No assumption that L is globally generated is required for this bound.

The resulting Hilbert polynomials must satisfy further characteristic-number identities. In odd dimensions we use $\chi(X, \Omega_X^p) = (-1)^p$ to eliminate Chern coefficients or Chern power sums. This produces a polynomial equation in an integer variable, with coefficients depending on the bounded section dimensions. In even dimensions we use the signature. The middle intersection form is (1) in the complex orientation, so $\text{sign}(X) = 1$. The Hilbert polynomial determines the Todd class and hence, after removing the factor $e^{c_1/2}$, the $\hat{A}$ -class. A characteristic power-series identity then recovers the signature and gives a polynomial equation in the section dimensions.

We exclude integer solutions of these equations by congruences and discriminant calculations. For $n = 7$ and $r = 4$, reduction modulo nine gives a contradiction. For $n = 7$ and $r = 2$, the discriminant is not a square for any of the 1224 allowed pairs of section dimensions. In the other computational cases, we fix the bounded parameters and reduce the polynomial in the remaining variable modulo

small primes or prime powers. For each parameter tuple, one of these reductions has no root. This excludes an integer root without requiring a bound on the remaining variable.

The assertions in arbitrary dimension also reduce to finitely many cases. Put $c = n + 1 - r$. Equation (1.1) implies

$$r \mid \frac{c^2(c-1)}{2}.$$

For each fixed positive c , only finitely many pairs (n, r) remain. We treat $c = 2, 4$, and also $c = 6, 8$ when n is even. Most of these pairs are excluded by the same Riemann–Roch or signature calculations. For $(n, r) = (15, 12), (27, 24)$, integrality of the twelfth Chern power sum, using the numerator 691 of B_{12} , forces $h^0(L) = n$ or $n - 1$ after the projective-space case is removed. General sections then cut out an integral Gorenstein curve of arithmetic genus two. Its section ring determines a weighted complete-intersection presentation of X . The Euler characteristic of either presentation differs from $n + 1$, giving the required contradiction.

The finite calculations are performed by the accompanying Python file `verify_fujita_further_extensions.py`, which we call the *verification program*. It derives the polynomial equations from the recurrences in the article and carries out the integer-square and congruence tests, using exact integer and rational arithmetic. Section 12 explains the implementation and its relation to the proofs. The source is distributed separately.

Sections 2–5 establish the numerical and characteristic-class formulas. Sections 6–9 treat dimensions seven through ten. The index bounds are proved in Sections 10 and 11. The appendices contain the longer polynomial expressions and the complete sieve tables. In dimension eleven, the restrictions above leave $r = 2, 4, 6, 8, 12$; our argument excludes 8, while 12 gives projective space. The cases $r = 2, 4, 6$ remain open in this approach.

Throughout, $\mathbb{P}^n$ denotes complex projective space. We use additive notation for tensor products of line bundles when convenient, so that $-K_X = rL$ means $K_X^{-1} \simeq L^{\otimes r}$. Characteristic classes and their evaluations are taken in the complex orientation.

2. NUMERICAL RESTRICTIONS

2.1. The ample generator. Because X is projective, it is Kähler. An ample divisor class and each of its powers are nonzero by positivity. Since $H^{2p}(X; \mathbb{C})$ has dimension one, that class spans its (p, p) part. Hodge decomposition therefore gives

$$h^{p,q}(X) = \begin{cases} 1 & p = q, \ 0 \leq p \leq n, \\ 0 & \text{otherwise.} \end{cases} \quad (2.1)$$

In particular $H^1(X, \mathcal{O}_X) = H^2(X, \mathcal{O}_X) = 0$. The exponential sequence implies

$$c_1 : \text{Pic}(X) \xrightarrow{\simeq} H^2(X; \mathbb{Z}) \simeq \mathbb{Z}.$$

One of the two primitive generators corresponds to a line bundle L with an ample positive power, and hence L itself is ample. Set $x = c_1(L)$. The ring assumption says that $\int_X x^n$ is either 1 or -1 ; positivity of an ample class in the complex orientation gives

$$\int_X x^n = 1. \quad (2.2)$$

All characteristic classes will be expressed using this choice. Write $c_j(T_X) = c_j x^j$, so that the symbols c_j on the right are integers. Since $-K_X$ is ample, its coefficient $r = c_1$ is a positive integer.

2.2. Holomorphic Euler characteristics. Equation (2.1) gives, for every p ,

$$\chi(X, \Omega_X^p) = \sum_q (-1)^q h^{p,q}(X) = (-1)^p. \quad (2.3)$$

It also gives $\chi_{\text{top}}(X) = n + 1$. The top Chern class is the Euler class of the underlying real tangent bundle, so

$$c_n = n + 1. \quad (2.4)$$

We derive the relation between $c_1 c_{n-1}$ and c_n from the χ_y -genus. Let $\alpha_1, \dots, \alpha_n$ be formal Chern roots, and put $\chi_y(X) = \sum_p \chi(X, \Omega_X^p) y^p$. Riemann–Roch gives

$$\chi_{-1+t}(X) = \int_X \prod_{i=1}^n \alpha_i \left(1 + \frac{t}{e^{\alpha_i} - 1} \right).$$

Use

$$\frac{1}{e^\alpha - 1} = \frac{1}{\alpha} - \frac{1}{2} + \frac{\alpha}{12} + \text{terms of degree at least three}.$$

In the coefficient of t^2 , only the terms with total root-degree zero after multiplication by $\prod_i \alpha_i$ contribute to the integral. They give

$$[t^2] \chi_{-1+t}(X) = \frac{1}{4} \binom{n}{2} c_n + \frac{1}{12} (c_1 c_{n-1} - n c_n) = \frac{n(3n-5)c_n}{24} + \frac{c_1 c_{n-1}}{12}.$$

To see the middle equality, expand $c_1 c_{n-1} = (\sum_j \alpha_j)(\sum_i \prod_{k \neq i} \alpha_k)$; the $i = j$ terms give $n c_n$. On the other hand, (2.3) gives

$$\chi_{-1+t}(X) = \sum_{p=0}^n (1-t)^p, \quad [t^2] \chi_{-1+t}(X) = \sum_{p=0}^n \binom{p}{2} = \binom{n+1}{3}.$$

Substitute $c_n = n + 1$ and solve:

$$r c_{n-1} = \frac{n(n+1)^2}{2}. \quad (2.5)$$

In particular, r divides the integer on the right.

2.3. The Hilbert polynomial and the index. Let $P(t) = \chi(X, L^t)$. Riemann–Roch gives a polynomial of degree n :

$$P(t) = \frac{t^n}{n!} + \frac{r t^{n-1}}{2(n-1)!} + \dots, \quad P(0) = 1. \quad (2.6)$$

The leading coefficient uses (2.2), and the next coefficient uses $\text{td}_1 = c_1/2$.

For an integer $t > -r$,

$$L^t = K_X \otimes L^{t+r}.$$

The second factor is ample, so Kodaira vanishing gives $H^i(X, L^t) = 0$ for $i > 0$. If $t < 0$, an anti-ample line bundle has no section: a nonzero section would give an effective divisor with negative intersection against L^{n-1} . It follows that

$$P(t) = h^0(X, L^t) \quad (t \geq 0), \quad P(-1) = \dots = P(-(r-1)) = 0. \quad (2.7)$$

There are $r-1$ distinct roots of a nonzero polynomial of degree n , hence

$$1 \leq r \leq n+1. \quad (2.8)$$

Serre duality also gives the polynomial identity

$$P(-r-t) = (-1)^n P(t). \quad (2.9)$$

The parity restriction can be obtained without any assertion about Pontryagin classes. The polynomial

$$P(t) - \binom{t+n}{n}$$

is integer-valued on integers and has degree at most $n-1$. Its leading coefficient is $(r-n-1)/(2(n-1)!)$. The $(n-1)$ st forward difference of an integer-valued polynomial of degree at most $n-1$ is its leading coefficient times $(n-1)!$, and is an integer. Thus

$$r \equiv n+1 \pmod{2}. \quad (2.10)$$

Here $\Delta f(t) = f(t+1) - f(t)$; applying it d times to at^d gives $d!a$.

2.4. The signature. If n is even, the middle real cohomology of the real $2n$ -manifold X is $H^n(X; \mathbb{R}) = \mathbb{R}x^{n/2}$. Its intersection matrix is the one-by-one matrix

$$\left(\int_X x^{n/2} x^{n/2} \right) = (1).$$

Consequently

$$\text{sign}(X) = 1. \quad (2.11)$$

Here the fundamental class is taken in the complex orientation, for which $\int_X x^n = 1$ by ampleness.

3. SPACES OF SECTIONS

3.1. The degree bound. We use the elementary inequality

$$\deg Z \geq N - d + 1 \quad (3.1)$$

for an integral nondegenerate d -dimensional variety $Z \subset \mathbb{P}^N$. Here nondegenerate means that Z is not contained in a hyperplane. One proof cuts by general hyperplanes. Such a section remains nondegenerate: from $H^0(\mathcal{O}_Z) = \mathbb{C}$ one has $H^1(\mathcal{I}_Z) = 0$, and the restriction sequence for the ideal shows that a new linear equation on a hyperplane section would have come from a linear equation on Z . General sections of positive dimension are integral by Bertini. After d cuts one has a zero-dimensional scheme of length $\deg Z$ spanning $\mathbb{P}^{N-d}$. The restrictions of its $N-d+1$ independent linear coordinates are independent sections of a line bundle on a scheme of that length; hence the length is at least $N-d+1$.

Lemma 3.1. *If A is ample on a smooth projective n -fold, then $h^0(X, A) \leq A^n + n$.*

Proof. The claim is immediate if $h^0(A) \leq 1$. Otherwise resolve the complete linear system by $\mu : Y \rightarrow X$ and write

$$\mu^* A = M + E,$$

where M is globally generated and E is effective. Both $\mu^* A$ and M are nef. Let d be the dimension of the image $Z \subset \mathbb{P}^{h^0(A)-1}$. Telescoping powers, each term of the difference $(\mu^* A)^n - (\mu^* A)^{n-d} M^d$ is an intersection of E with nef classes; therefore it is nonnegative. The projection formula gives

$$A^n \geq (\mu^* A)^{n-d} M^d \geq \deg Z.$$

The middle intersection is $\deg Z$ times a positive integer: the $\mu^* A$ -degree of a general fiber, or the generic degree if $d = n$. A general fiber meets the locus where μ is an isomorphism, so its image has the same dimension and positive degree with respect to A . By (3.1), $\deg Z \geq h^0(A) - d$. Since $d \leq n$, the result follows. $\square$

Applying this to $A = L^k$ gives

$$0 \leq h^0(X, L^k) \leq k^n + n \quad (k \geq 1). \quad (3.2)$$

Lemma 3.2. *Suppose $L^n = 1$ and $m = h^0(X, L) > 0$. The image of $|L|$ is the full $\mathbb{P}^{m-1}$. If the base scheme is nonempty, its codimension is m .*

Proof. With $A = \mu^*L$, the proof above gives

$$1 = A^n \geq A^{n-d} M^d \geq \deg Z \geq m - d \geq 1.$$

Every inequality is an equality. In particular, $d = m - 1$, so the closed image in $\mathbb{P}^{m-1}$ is all of that space.

Let the base scheme have codimension $b < m$. Choose b general members of $|L|$. They intersect properly: before the b th cut the base scheme cannot force a component of smaller codimension, and general sections avoid all other components at each stage. Their intersection cycle contains a base-locus component. It also contains a component outside the base locus, because b general hyperplanes intersect the dense open image in $\mathbb{P}^{m-1}$. These are distinct effective components, each with positive integral L -degree. Their total L -degree, however, is $L^n = 1$. This is impossible. Thus $b \geq m$. An ideal generated by the m sections has height at most m , which gives $b = m$. $\square$

Lemma 3.3. *If $h^0(X, L) = n + 1$, then $(X, L) \simeq (\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1))$. Also, if $r = n + 1$, then $X \simeq \mathbb{P}^n$.*

Proof. The preceding lemma makes the base scheme empty when $m = n + 1$. The resulting map $f : X \rightarrow \mathbb{P}^n$ has $L = f^*\mathcal{O}(1)$. Ampleness of L precludes a positive-dimensional fiber, since any curve in such a fiber would have L -degree zero. Thus f is finite. Its degree is $L^n = 1$. A finite birational map to a normal variety is an isomorphism.

If $r = n + 1$, the n roots in (2.7) and the leading coefficient of P determine

$$P(t) = \frac{(t+1) \cdots (t+n)}{n!}.$$

Then $h^0(L) = P(1) = n + 1$ and the first assertion applies. $\square$

The second assertion is the equality case of the Kobayashi–Ochiai characterization [KO73]; the proof above uses the additional hypothesis $L^n = 1$. For $r < n + 1$, the first assertion gives $0 \leq h^0(L) \leq n$. The weaker bound $n + 1$ will suffice in dimensions seven and eight.

4. CHARACTERISTIC CLASSES AND RIEMANN–ROCH

4.1. Formal roots and Newton identities. Let s_j be the coefficient of x^j in the j th power sum of the Chern roots, with $s_0 = n$. Thus $s_1 = r$. The s_j are symmetric polynomials in the formal Chern roots. Newton's identities give

$$c_k = \frac{1}{k} \sum_{j=1}^k (-1)^{j-1} s_j c_{k-j}, \quad c_0 = 1. \quad (4.1)$$

Conversely,

$$s_k = \sum_{j=1}^{k-1} (-1)^{j-1} c_j s_{k-j} + (-1)^{k-1} k c_k.$$

The latter formula shows inductively that every s_k is an integer. The first examples are $s_2 = r^2 - 2c_2$ and $s_3 = r^3 - 3rc_2 + 3c_3$.

In all series below, the formal variable x records *complex* degree: x^j represents a cohomology class of real degree $2j$. Calculations are truncated after x^n .

4.2. The Todd class. With Bernoulli numbers defined by $z/(e^z - 1) = \sum B_j z^j / j!$,

$$\log \mathrm{td}(T_X) = \frac{rx}{2} - \sum_{2j \leq n} \frac{B_{2j} s_{2j}}{2j(2j)!} x^{2j}. \quad (4.2)$$

Only even Bernoulli numbers are used. For example, through degree eight,

$$\log \mathrm{td} = \frac{rx}{2} - \frac{s_2 x^2}{24} + \frac{s_4 x^4}{2880} - \frac{s_6 x^6}{181440} + \frac{s_8 x^8}{9676800}.$$

These coefficients come from summing $\log(\alpha/(1 - e^{-\alpha}))$ over the formal roots.

We shall use the following recurrences for truncated power series. If $A(x) = 1 + \sum_{j \geq 1} a_j x^j$ and $\log A(x) = \sum_{j \geq 1} b_j x^j$, then

$$b_j = a_j - \frac{1}{j} \sum_{k=1}^{j-1} k b_k a_{j-k}. \quad (4.3)$$

If $A(x) = \exp B(x)$, the converse is

$$a_0 = 1, \quad a_j = \frac{1}{j} \sum_{k=1}^j k b_k a_{j-k}. \quad (4.4)$$

Both follow from $A' = B'A$. Products are computed by the usual finite convolution $[x^j](UV) = \sum_{k=0}^j u_k v_{j-k}$. In particular, all these operations take place over $\mathbb{Q}$.

4.3. Exterior powers of the cotangent bundle. If $\text{td}(T_X) = \sum_{j=0}^n \text{td}_j x^j$, Riemann–Roch is

$$P(t) = [x^n] e^{tx} \text{td}(T_X), \quad \text{td}_j = (n-j)! [t^{n-j}] P(t). \quad (4.5)$$

The Chern character of the cotangent bundle is

$$U(x) = \text{ch}(\Omega_X^1) = \sum_{j=0}^n \frac{(-1)^j s_j}{j!} x^j.$$

Writing $U_p(x) = \text{ch}(\Lambda^p \Omega_X^1)$, with $U_0 = 1$, Newton's exterior-power identity is

$$p U_p(x) = \sum_{a=1}^p (-1)^{a-1} U_{p-a}(x) U(ax). \quad (4.6)$$

Indeed, the roots of the exterior-power generating function are $e^{-\alpha_1}, \dots, e^{-\alpha_n}$; their a th power sum is $U(ax)$. For example, $U_2 = (U(x)^2 - U(2x))/2$. Finally,

$$[x^n] U_p(x) \text{td}(T_X) = \chi(X, \Omega_X^p) = (-1)^p. \quad (4.7)$$

We will use these recurrences to express the Riemann–Roch identities as polynomial equations in the Chern power sums.

5. THE HILBERT SERIES AND THE SIGNATURE

5.1. Reciprocity of the Hilbert numerator. Put $H(z) = \sum_{k \geq 0} P(k) z^k$. Finite differences of a degree- n polynomial show that

$$H(z) = \frac{h(z)}{(1-z)^{n+1}}, \quad h(z) \in \mathbb{Z}[z], \quad \deg h \leq n.$$

For example, the coefficient of z^j , $j \geq n+1$, in $(1-z)^{n+1} H(z)$ is the $(n+1)$ st difference of P , and is zero. We have $h(0) = P(0) = 1$. The leading coefficient $1/n!$ of P gives $h(1) = 1$ by comparison of the leading pole at $z = 1$.

The rational generating function of a polynomial satisfies

$$H(z^{-1}) = - \sum_{k \geq 1} P(-k) z^k.$$

The equality means expansion of the rational function on the left at zero. It follows first for $P(t) = 1$ from $(1-z^{-1})^{-1} = -z/(1-z)$; applying $z d/dz$ proves it for monomials, and hence for every polynomial. Using (2.7) and (2.9) on the right gives

$$H(z^{-1}) = (-1)^{n+1} z^r H(z).$$

Substituting the rational form of H and cancelling the denominators yields

$$h(z) = z^{n+1-r} h(z^{-1}). \quad (5.1)$$

Since $h(0) = 1$, the degree is exactly $n + 1 - r$, its leading coefficient is one, and its coefficients are symmetric. By (2.10), write $n + 1 - r = 2s$.

Let $v_i = h^0(X, L^i)$, with $v_0 = 1$, and $h(z) = \sum_{j=0}^{2s} h_j z^j$. Coefficient comparison gives

$$\begin{aligned} h_i &= v_i - \sum_{j=0}^{i-1} h_j \binom{n+i-j}{n} \quad (0 \leq i < s), \\ h_s &= 1 - 2 \sum_{j=0}^{s-1} h_j, \quad h_{2s-j} = h_j. \end{aligned} \tag{5.2}$$

Thus only $v_1, \dots, v_{s-1}$ are free parameters. We make no assertion that the coefficients h_j are nonnegative: L need not generate its section ring in degree one.

5.2. Recovering the Todd class. The following identity holds modulo x^{n+1} :

$$\mathrm{td}(T_X) = \left(\frac{x}{1 - e^{-x}} \right)^{n+1} h(e^{-x}). \tag{5.3}$$

The binomial coefficient identity follows by a formal residue calculation. For any integer j ,

$$[x^n] e^{(t-j)x} \left(\frac{x}{1 - e^{-x}} \right)^{n+1} = \mathrm{Res}_{x=0} \frac{e^{(t-j)x} dx}{(1 - e^{-x})^{n+1}}.$$

Put $z = 1 - e^{-x}$, so $dx = dz/(1 - z)$. The residue becomes $[z^n](1 - z)^{-(t-j+1)} = \binom{t+n-j}{n}$. Multiplication by h_j and summation gives exactly the Hilbert polynomial. Since (4.5) recovers each coefficient separately, the truncated classes are equal.

5.3. The signature formula. Set

$$q(x) = \frac{x}{2 \sinh(x/2)}, \quad B(x) = e^{sx} h(e^{-x}).$$

Reciprocity makes B even, and (5.3) gives

$$\hat{A}(T_X) = e^{-rx/2} \mathrm{td}(T_X) = q(x)^{n+1} B(x).$$

This is an identity of rational characteristic classes; it requires no spin structure. The characteristic power series obey

$$\frac{q(2x)^2}{q(4x)} = \frac{x}{\tanh x}.$$

Consequently, root by root and then by multiplication, the Hirzebruch L -class is

$$\mathcal{L}(T_X) = \left(\frac{x}{\tanh x} \right)^{n+1} \frac{B(2x)^2}{B(4x)}.$$

The factors e^{2sx} and e^{4sx} cancel in this quotient. Taking the top coefficient and making the formal change of variable $t = \tanh x$ gives

$$\mathrm{sign}(X) = [t^n] \frac{h\left(\frac{1-t}{1+t}\right)^2}{(1-t^2)h\left(\left(\frac{1-t}{1+t}\right)^2\right)}. \tag{5.4}$$

Indeed, the coefficient is initially the residue of $h(e^{-2x})^2 dx / (\tanh(x)^{n+1} h(e^{-4x}))$. Now $dx = dt/(1-t^2)$ and $e^{-2x} = (1-t)/(1+t)$.

To express this in terms of polynomials, put $d = 2s$ and define

$$N(t) = \sum_{j=0}^d h_j (1-t)^j (1+t)^{d-j}, \quad (5.5)$$

$$D(t) = (1-t^2) \sum_{j=0}^d h_j (1-t)^{2j} (1+t)^{2d-2j}.$$

Both are even, and $N(0) = D(0) = 1$. Thus, in even complex dimension,

$$[t^n] \frac{N(t)^2}{D(t)} = 1. \quad (5.6)$$

If $D(t) = 1 + \sum_{j \geq 1} d_j t^{2j}$ and $N(t)^2 = \sum_{j \geq 0} b_j t^{2j}$, set

$$a_0 = b_0 = 1, \quad a_j = b_j - \sum_{i=1}^j d_i a_{j-i}. \quad (5.7)$$

Then (5.6) is $a_{n/2} = 1$. Every coefficient in this recurrence is an integer polynomial in the section counts.

5.4. Reduction modulo an integer. Let $F \in \mathbb{Z}[y]$. If F has an integer root, its reduction modulo any positive integer q has a root in $\mathbb{Z}/q\mathbb{Z}$. We shall use the contrapositive: a modulus for which the reduction has no root excludes all integer roots of F . When the coefficients depend on a bounded integer tuple $\mathbf{v}$, we may choose a different modulus for each specialization $F_{\mathbf{v}}$.

We first clear denominators and divide out the greatest common divisor of the coefficients over $\mathbb{Q}$, then reduce the resulting integer polynomial modulo q . These operations preserve its rational zero set. In particular, a prime dividing a denominator in the derivation may still be used to exclude integer roots of the resulting polynomial.

6. DIMENSION SEVEN

6.1. The possible indices. In this section only, write the Chern coefficients as

$$(c_1, c_2, c_3, c_4, c_5, c_6, c_7) = (r, a, b, c, d, e, f).$$

In particular, $b = c_3$; the power sum s_3 retains the notation of Section 4. Equations (2.4)–(2.10) become

$$f = 8, \quad re = 224, \quad 1 \leq r \leq 8, \quad r \text{ even}.$$

Among 2, 4, 6, 8, the value 6 does not divide 224. Hence

$$r \in \{2, 4, 8\}. \quad (6.1)$$

The value 8 gives $\mathbb{P}^7$ by Lemma 3.3. It remains to exclude 4 and 2.

6.2. The Chern-number equations. Apply the recurrences of Section 4, substituting $e = 224/r$ and $f = 8$. Let

$$\rho_0 = \chi(\mathcal{O}_X) - 1, \quad \rho_1 = \chi(\Omega_X^1) + 1, \quad \rho_2 = \chi(\Omega_X^2) - 1.$$

The resulting identities are

$$E_1 = -7200\rho_0 + 1440\rho_1, \quad E_2 = -120960\rho_0, \quad \rho_2 = -6\rho_0 + 3\rho_1,$$

where

$$E_1 = r^3c - 3rac - r^2d + 3bc - 3ad + 7728, \quad (6.2)$$

$$\begin{aligned} E_2 = & -2r^5a + 10r^3a^2 + 2r^4b - 10ra^3 - 11r^2ab \\ & - 2r^3c + rb^2 + 9rac + 2r^2d + 120512. \end{aligned} \quad (6.3)$$

Under the substitutions $e = 224/r$ and $f = 8$, the three Euler-characteristic identities are therefore equivalent to $E_1 = E_2 = 0$.

For use in the elimination, the Hilbert polynomial is

$$\begin{aligned}
60480P(t) = & 12t^7 + 42rt^6 + 42(r^2 + a)t^5 + 105rat^4 \\
& + 14(-r^4 + 4r^2a + 3a^2 + rb - c)t^3 \\
& + 21(-r^3a + 3ra^2 + r^2b - rc)t^2 \\
& + (2r^6 - 12r^4a + 11r^2a^2 + 5r^3b + 10a^3 \\
& + 11rab - 5r^2c - b^2 - 9ac - 2rd + 2e)t + 60480.
\end{aligned} \tag{6.4}$$

The constant term uses $\rho_0 = 0$. The remaining coefficients follow from (4.2), (4.4), and (4.5), truncated in degree seven. For example, the coefficient of t^5 is $\text{td}_2/5! = (r^2 + a)/(12 \cdot 5!)$, which is $42(r^2 + a)/60480$ as displayed.

6.3. The case $r = 4$. Set $r = 4$, $e = 56$, and $m = P(1)$. The coefficient of d in E_2 is $2r^2 = 32$, so solve $E_2 = 0$ uniquely for d . Substitution in (6.4) gives

$$P(-1) = -\frac{3a^2 - 41a + 4b - c - 598}{1440}, \quad P(1) = \frac{3a^2 - 17a + 4b - c + 458}{288}.$$

Kodaira vanishing gives $P(-1) = 0$. Subtract the corresponding numerators: $288m = 24a + 1056$, hence

$$a = 12m - 44, \quad c = 4b + 432m^2 - 3660m + 7014.$$

The solved equation for d becomes

$$8d = -b^2 + 96bm - 352b - 29376m^3 + 367056m^2 - 1451928m + 1787736.$$

Substituting these expressions in $2E_1 = 0$ gives

$$\begin{aligned}
F_4(m, b) = & (9m - 5)b^2 + (1728m^2 - 17160m + 36612)b \\
& + 264384m^4 - 4279824m^3 + 25277544m^2 \\
& - 64549008m + 60164376 = 0.
\end{aligned} \tag{6.5}$$

Only nonzero constants have been divided out in these substitutions.

Reduce all coefficients modulo nine:

$$F_4(m, b) \equiv 4b^2 + 3bm + 6 \pmod{9}.$$

Reducing further modulo three forces $b^2 \equiv 0 \pmod{3}$, so $3 \mid b$. Then both $4b^2$ and $3bm$ are divisible by nine, while 6 is not. This contradiction excludes $r = 4$, for every integer m without any bound.

6.4. The case $r = 2$. Now $r = 2$ and $e = 112$. Set $m = P(1)$ and $k = P(2)$. Equation (3.2) gives

$$0 \leq m \leq 8, \quad 0 \leq k \leq 2^7 + 7 = 135. \tag{6.6}$$

Again solve $E_2 = 0$ for d , this time dividing by the nonzero constant 8. The two evaluations of P reduce to

$$P(1) = \frac{3a^2 + a + 2b - c + 1442}{720}, \quad P(2) = \frac{3a^2 + 16a + 2b - c + 572}{180}.$$

Thus $180k - 720m = 15a - 870$. Put $u = k - 4m$; then

$$\begin{aligned}
a = & 12u + 58, \\
c = & 2b + 432u^2 + 4188u + 11592 - 720m, \\
4d = & -b^2 + 48bu + 232b + 77760mu + 370080m \\
& - 29376u^3 - 429552u^2 - 2248824u - 4200128.
\end{aligned}$$

With these substitutions, $2E_1 = 0$ becomes

$$Ab^2 + Bb + C = 0, \tag{6.7}$$

where

$$\begin{aligned}
A &= 18u + 101, \\
B &= 1728u^2 + 16392u + 47544 - 4320m, \\
C &= 528768u^4 + 10284192u^3 + 77812128u^2 \\
&\quad + 271230552u + 365944288 \\
&\quad - m(1399680u^2 + 13478400u + 32447520).
\end{aligned}$$

The integer A cannot vanish, since 101 is not divisible by 18. An integral root b would give the identity

$$B^2 - 4AC = (2Ab + B)^2. \quad (6.8)$$

Therefore the discriminant must be a nonnegative integer square.

The rectangle (6.6) contains $9 \cdot 136 = 1224$ pairs. Computing u, A, B, C and $D = B^2 - 4AC$ for each pair gives

m	0	1	2	3	4	5	6	7	8	total
$D \geq 0$	0	0	4	7	12	16	20	25	30	114
D square	0	0	0	0	0	0	0	0	0	0

For each nonnegative discriminant, we compute the integer $\lfloor \sqrt{D} \rfloor$ and compare its square with D . Alternatively, requiring the 114 nonnegative discriminants to be quadratic residues successively modulo 5, 7, 11, 13, 17, 19, 23, 29 leaves 90, 60, 33, 23, 16, 8, 3, 0 pairs. Either calculation excludes every pair in (6.6).

Both nonmaximal indices in (6.1) are excluded. The remaining value $r = 8$ gives $X \simeq \mathbb{P}^7$ by Lemma 3.3. This proves C_7 .

7. DIMENSION EIGHT

7.1. The possible indices. Now $c_8 = 9$ and $rc_7 = 324$. Bounds and parity give the candidates 1, 3, 5, 7, 9; divisibility by 324 removes 5, 7. Hence

$$r \in \{1, 3, 9\}. \quad (7.1)$$

The value 9 gives $\mathbb{P}^8$ by Lemma 3.3.

7.2. The Hilbert polynomial. Since n is even, (2.9) says $P(t) = P(-r - t)$. Put $y = t + r/2$. Invariance under $y \mapsto -y$ means that P is a polynomial in y^2 , equivalently in $z = t(t + r) = y^2 - r^2/4$. The leading coefficient and $P(0) = 1$ give

$$P(t) = \frac{z^4 + Az^3 + Bz^2 + Cz + 40320}{40320}, \quad z = t(t + r). \quad (7.2)$$

The letters A, B, C in this section denote Hilbert-polynomial coefficients.

Set $m = P(1)$, $k = P(2)$, $\ell = P(3)$. The bounds needed are

$$0 \leq m \leq 9, \quad 0 \leq k \leq 2^8 + 8 = 264. \quad (7.3)$$

For $r = 3$ the values of z at $t = -1, 1, 2$ are $-2, 4, 10$. The three equations for A, B, C are therefore

$$\begin{aligned}
-8A + 4B - 2C &= -40336, \\
64A + 16B + 4C &= 40320m - 40576, \\
1000A + 100B + 10C &= 40320k - 50320.
\end{aligned}$$

Their coefficient matrix is invertible, as is also clear from polynomial interpolation at three distinct nonzero values of z . Solving gives

$$\begin{aligned}
A &= 56k - 280m + 492, \\
B &= -112k + 2240m - 6036, \\
C &= -448k + 5600m + 6128.
\end{aligned} \quad (7.4)$$

For $r = 1$, the values of z at $t = 1, 2, 3$ are 2, 6, 12. The equations are

$$\begin{aligned} 8A + 4B + 2C &= 40320m - 40336, & 216A + 36B + 6C &= 40320k - 41616, \\ 1728A + 144B + 12C &= 40320\ell - 61056. \end{aligned}$$

Their solution is

$$\begin{aligned} A &= -280k + 56\ell + 504m - 300, \\ B &= 3920k - 448\ell - 9072m + 5708, \\ C &= -6720k + 672\ell + 36288m - 30384. \end{aligned} \tag{7.5}$$

7.3. The signature equation. In dimension eight we can write the signature in terms of the four coefficients of the $\hat{A}$ -class. By (4.5), $\text{td}_j = (8-j)! [t^{8-j}]P(t)$. Write

$$e^{-rx/2} \sum_{j=0}^8 \text{td}_j x^j = 1 + Ux^2 + Vx^4 + Wx^6 + Zx^8.$$

The odd coefficients vanish, either by (2.9) or by the Todd power-series identity. The logarithm has coefficients

$$\begin{aligned} q_1 &= U, \\ q_2 &= V - U^2/2, \\ q_3 &= W - UV + U^3/3, \\ q_4 &= Z - UW - V^2/2 + U^2V - U^4/4. \end{aligned}$$

The conversion to the logarithm of the L -class is:

	$\log \hat{A}$ coefficient of s_{2j}	$\log \mathcal{L}$ coefficient of s_{2j}	ratio
$j = 1$	$-1/24$	$1/3$	-8
$j = 2$	$1/2880$	$-7/90$	-224
$j = 3$	$-1/181440$	$62/2835$	-3968
$j = 4$	$1/9676800$	$-127/18900$	-65024

Each entry follows by expanding $\log(x/(2 \sinh(x/2)))$ or $\log(x/\tanh x)$ and summing over roots. Set $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (-8q_1, -224q_2, -3968q_3, -65024q_4)$. Exponentiating and extracting degree eight gives

$$\text{sign}(X) = \lambda_4 + \lambda_1\lambda_3 + \frac{\lambda_2^2}{2} + \frac{\lambda_1^2\lambda_2}{2} + \frac{\lambda_1^4}{24}. \tag{7.6}$$

By (2.11), this number must be 1.

7.4. The case $r = 3$. Substitute (7.4) into (7.2) and then into (7.6). Set $u = k - 5m$. We obtain the polynomial identity $\text{sign}(X) - 1 = 32F_3(m, u)$, where

$$\begin{aligned} F_3(m, u) &= 1152u^4 + 40920u^3 + (557147 - 4032m)u^2 \\ &\quad + (3443306 - 71232m)u + 1800m^2 - 325410m + 8148917. \end{aligned} \tag{7.7}$$

Thus $F_3(m, u) = 0$ for an integer u and $m = 0, \dots, 9$. For each value of m , a modulus excluding every residue of u is:

m	0	1	2	3	4	5	6	7	8	9
q	11	11	7	7	11	7	7	19	11	7

For each entry, $F_3(m, u)$ is nonzero modulo q for $u = 0, \dots, q-1$. Hence there is no integer root for any of the allowed values of m , and $r = 3$ is impossible. The argument uses no bound on u or k .

7.5. **The case $r = 1$.** For (7.5), set $u = \ell - 5k + 9m$ and $v = k - 3m$. The same calculation now gives $\text{sign}(X) - 1 = 32F_1(m, k, u)$, where

$$\begin{aligned} F_1(m, k, u) = & 1152u^4 - 23592u^3 + 173099u^2 - 537930u + 596995 \\ & + 1800v^2 + (-4032u^2 + 41664u - 100434)v \\ & + (3024u - 16000)m, \quad v = k - 3m. \end{aligned} \quad (7.8)$$

The two bounds in (7.3) give $10 \cdot 265 = 2650$ pairs (m, k) . For each pair, treat (7.8) as a polynomial in the unrestricted integer u . The successive congruence tests give the following counts:

q	remaining pairs	q	remaining pairs
initial	2650	13	39
4	1325	17	25
9	662	19	14
16	332	23	8
25	160	29	3
27	148	31	2
49	80	37	1
11	64	43	0

The moduli are applied down the left column and then down the right. As explained in Section 5, an integer root must also give a root modulo each composite modulus. No pairs remain, so $r = 1$ is impossible for every integer ℓ . Equation (7.1) leaves $r = 9$, and Lemma 3.3 gives $X \simeq \mathbb{P}^8$. This proves C_8 .

8. DIMENSION NINE

The basic restrictions give $r \mid 450$, r even, and $r \leq 10$. Thus $r = 2, 6, 10$. The value 10 gives projective space. The value 6 is excluded in Section 10.2 by direct characteristic-number equations; that argument does not assume C_9 . We treat $r = 2$ here.

8.1. **Section dimensions and even power sums.** Set $m = h^0(L)$, $k = h^0(2L)$, and $\ell = h^0(3L)$. Lemma 3.3 excludes $m = 10$ in the present index, so

$$0 \leq m \leq 9, \quad 0 \leq k \leq 2^9 + 9 = 521, \quad 0 \leq \ell \leq 3^9 + 9 = 19692. \quad (8.1)$$

Since $n + 1 - r = 8$, the first half of $h(z)$ is

$$h_0 = 1, \quad h_1 = m - 10, \quad h_2 = k - 10m + 45, \quad h_3 = \ell - 10k + 45m - 120.$$

Set $h_4 = 1 - 2(h_0 + h_1 + h_2 + h_3)$ and reflect the coefficients.

In this section put

$$u = s_2, \quad v = s_4, \quad w = s_6, \quad a = s_3, \quad b = s_5.$$

Thus the letters a, b in this section denote odd Chern power sums. Let $B(x) = e^{4x}h(e^{-x})$. If

$$T = \ell - 6k + 14m - 14, \quad Q = \ell + 6k - 34m + 46, \quad J = \ell + 54k + 134m - 434,$$

direct expansion of the paired exponentials gives

$$[x^2]B = T, \quad [x^4]B = Q/12, \quad [x^6]B = J/360.$$

For example, $B = 2 \cosh 4x + 2h_1 \cosh 3x + 2h_2 \cosh 2x + 2h_3 \cosh x + h_4$, so its degree-two coefficient is $16 + 9h_1 + 4h_2 + h_3 = T$.

Since $\log \hat{A} = 10 \log q + \log B$, formula (4.3) gives

$$\begin{aligned} u &= 10 - 24T, \\ v &= 10 + 240Q - 1440T^2, \\ w &= 10 - 504J + 15120TQ - 60480T^3. \end{aligned} \quad (8.2)$$

For instance, $[x^4] \log B = Q/12 - T^2/2$, and the coefficient of s_4 in $\log q$ is $1/2880$.

8.2. Elimination of the odd power sums. Start from

$$(s_0, \dots, s_9) = (9, 2, u, a, v, b, w, s_7, s_8, s_9).$$

The equation $[x^9] \text{td} = 1$ is linear in s_8 with coefficient $1/9676800$, hence has the unique solution

$$\begin{aligned} s_8 = & -\frac{175}{144}u^4 + \frac{175}{9}u^3 - \frac{35}{12}u^2v - 70u^2 \\ & + \frac{70}{3}uv - \frac{20}{9}uw + 80u - \frac{7}{12}v^2 \\ & - 28v + \frac{80}{9}w + \frac{29030320}{3}. \end{aligned} \tag{8.3}$$

Indeed, $s_8 x^8 / 9676800$ in $\log \text{td}$ multiplies the degree-one term x of td .

Newton's identities now compute c_8 and c_9 . They must be

$$c_8 = \frac{9 \cdot 10^2}{2 \cdot 2} = 225, \quad c_9 = 10.$$

The coefficient of s_7 in c_8 is $s_1/7 = 2/7$. Thus, if $c_8^{(0)}$ denotes the Newton expression with $s_7 = 0$,

$$s_7 = \frac{7}{2}(225 - c_8^{(0)}).$$

After this substitution the coefficient of s_9 in c_9 is $1/9$, so, with the analogous notation, $s_9 = 9(10 - c_9^{(0)})$. These are linear eliminations with nonzero constant coefficients. They are therefore valid for every choice of u, v, w, a, b .

Impose (4.7) for $p = 1, 2$. Clearing denominators and coefficient gcds gives two integer polynomials

$$E_1(u, v, w, a, b) = 0, \quad E_2(u, v, w, a, b) = 0.$$

Each has degree one in b . Write $E_i = L_i b + C_i$. Their resultant is $L_1 C_2 - L_2 C_1$. Any common solution makes this expression zero, even if one or both coefficients L_i vanish. No division by L_i is needed.

Remove the resultant's nonzero coefficient gcd. The resulting polynomial $F(u, v, w, a)$ has 60 monomials and degree four in a :

$$F(u, v, w, a) = \sum_{i=0}^4 A_i(u, v, w) a^i, \quad A_4 = 3840u - 63488. \tag{8.4}$$

The equations E_1, E_2 and the five coefficients A_i are given in Appendix A. After substituting (8.2), the integer $a = s_3$ must satisfy (8.4).

8.3. The congruence calculation. The rectangle (8.1) contains $10 \cdot 522 \cdot 19693 = 102797460$ triples. For each triple, the integer polynomial in a must have a root modulo every prime. Test the primes from 13 through 223:

primes tested through	remaining triples
23	20583262
97	16021
149	136
181	3
223	0

Appendix D contains every intermediate count. The representation of the candidate set used for this calculation is described in Section 12. The empty final set excludes $r = 2$. Only $r = 10$ remains, proving C_9 .

9. DIMENSION TEN

Here $r \mid 605$, r is odd, and $r \leq 11$. The three possibilities are $r = 1, 5, 11$. Section 11.1 excludes $r = 5$ by a signature calculation independent of the present argument. The value 11 gives projective space. It remains to exclude $r = 1$.

9.1. The signature polynomial. Put $m = h^0(L)$, $k = h^0(2L)$, $\ell = h^0(3L)$, and $j = h^0(4L)$. Now h has degree ten, with

$$\begin{aligned} h_0 &= 1, & h_1 &= m - 11, \\ h_2 &= k - 11m + 55, & h_3 &= \ell - 11k + 55m - 165, \\ h_4 &= j - 11\ell + 55k - 165m + 330. \end{aligned}$$

Its central coefficient and its reflected coefficients are given by (5.2). Formula (5.7) gives

$$\text{sign}(X) - 1 = 8F_{10,1}(m, k, \ell, j), \quad (9.1)$$

where $F_{10,1}$ is a primitive integer polynomial with 126 monomials and degree five in j . Its leading coefficient in j is -73728 .

The polynomial can be expressed more briefly in terms of the following linear combinations:

$$\begin{aligned} T &= j - 7\ell + 20k - 28m + 14, \\ Q &= j + 5\ell - 40k + 80m - 46, \\ J &= j + 53\ell + 80k - 568m + 434, \\ V &= j + 245\ell + 3800k + 7280m - 11326. \end{aligned}$$

They are respectively $[x^2]B$, $12[x^4]B$, $360[x^6]B$, and $20160[x^8]B$ for $B(x) = e^{5x}h(e^{-x})$. Let

$$\begin{aligned} G(T, Q, J, V) &= -23224320T^5 + 5322240T^4 - 532224T^3 \\ &\quad + 30162T^2 + 12151T - 41207040 \\ &\quad + 8709120QT^3 - 1552320QT^2 + 110880QT - 22533Q \\ &\quad - 655200Q^2T + 57750Q^2 \\ &\quad - 233856JT^2 + 38808JT + 3654J + 17640JQ \\ &\quad + 3060TV - 1147V. \end{aligned} \quad (9.2)$$

Then

$$F_{10,1}(m, k, \ell, j) = \frac{G(T, Q, J, V)}{315}. \quad (9.3)$$

After the displayed linear substitutions, all coefficients of G are divisible by 315. The resulting integer polynomial agrees with that obtained from (5.7), and also with the expression recovered from the Todd class. Both identities are checked by the verification program.

The congruence tests are applied to $F_{10,1}$. The division by 315 is performed coefficientwise over $\mathbb{Q}$ before reduction modulo a prime, as in Section 5.

9.2. The congruence calculation. The bounds required for the first three section counts are

$$0 \leq m \leq 10, \quad 0 \leq k \leq 2^{10} + 10 = 1034, \quad 0 \leq \ell \leq 3^{10} + 10 = 59059.$$

The value $m = 11$ has been removed by Lemma 3.3. No bound on j is needed, since an integer root gives a root modulo every prime.

There are $11 \cdot 1035 \cdot 59060 = 672398100$ triples. The congruence tests give

primes tested through	remaining triples
31	8661101
97	14926
149	148
181	7
239	0.

Thus $F_{10,1}$ has no integer root in j for any allowed triple (m, k, ℓ) . This excludes $r = 1$, and only $r = 11$ remains. Hence $X \simeq \mathbb{P}^{10}$.

10. THE INDEX BOUND $r \geq n - 3$

10.1. The possible dimensions. Put $c = n + 1 - r$, an even nonnegative integer. Substituting $n = r + c - 1$ gives

$$\frac{n(n+1)^2}{2} - \frac{c^2(c-1)}{2} = \frac{r}{2}(r^2 + (3c-1)r + 3c^2 - 2c).$$

The bracket is even: $r^2 + (3c-1)r$ is even because $3c-1$ is odd, and $3c^2 - 2c$ is even. Equation (2.5) therefore implies

$$r \mid \frac{c^2(c-1)}{2}. \quad (10.1)$$

For $r \geq n - 3$, parity leaves $r = n + 1, n - 1, n - 3$. The first gives projective space. For $r = n - 1$, $c = 2$ and (10.1) gives $r \mid 2$, hence $n = 2, 3$, covered by the established low-dimensional result. For $r = n - 3$, $c = 4$ and $r \mid 24$, so the full list is

$$(n, r) = (4, 1), (5, 2), (6, 3), (7, 4), (9, 6), (11, 8), (15, 12), (27, 24). \quad (10.2)$$

The first three are again covered by $n \leq 6$. We first exclude dimensions 7, 9, 11 by Riemann–Roch, then treat 15, 27 using the geometry of a curve section.

10.2. Dimensions seven, nine, and eleven. Let $m = h^0(L)$, with $0 \leq m \leq n$, and $r = n - 3$. The Kodaira roots divide $P(t)$:

$$P(t) = \frac{\prod_{j=1}^{r-1}(t+j)}{n!}(z^2 + \alpha z + \beta), \quad z = t(t+r).$$

To justify the shape, the quotient has degree four. Under $t \mapsto -r - t$ both P and the product of roots pick up the same sign because $n - (r - 1) = 4$ is even. The quotient is invariant, so it is a quadratic polynomial in z ; its leading coefficient is one. The conditions $P(0) = 1$, $P(1) = m$ give

$$\beta = r(r+1)(r+2)(r+3), \quad \alpha = (r+2)(r+3)(m-r) - (r+1). \quad (10.3)$$

For example, the product at $t = 1$ is $r!$ and $z = r + 1$; substitution verifies the second formula directly.

Recover all Todd coefficients by (4.5), form $e^{-rx/2} \text{td}$, and take its logarithm. Dividing the coefficient of x^{2j} by $-B_{2j}/(2j(2j)!)$ determines s_{2j} . Set $s_1 = r$ and temporarily leave $s_3, s_5, \dots, s_n$ unknown.

The coefficient of s_{n-2} in c_{n-1} is $r/(n-2)$. Indeed, the generating identity

$$c(t) = \exp \left(\sum_{j \geq 1} (-1)^{j-1} s_j t^j / j \right)$$

gives $\partial c_{n-1} / \partial s_{n-2} = c_1 / (n-2)$ in these odd dimensions. Thus $c_{n-1} = n(n+1)^2 / (2r)$ determines s_{n-2} uniquely. Next, $c_n = n + 1$ determines s_n with coefficient $1/n$. The free odd variables left are

n	remaining variables
7	s_3
9	s_3, s_5
11	s_3, s_5, s_7 .

Use (4.6)–(4.7) for $p = 1, \dots, (n-5)/2$. This gives one, two, or three polynomial equations.

We compute a Gröbner basis over $\mathbb{Q}$, ordering the remaining variables lexicographically from the largest subscript down to s_3 . The basis contains a nonzero polynomial solely in s_3 . After clearing denominators and a coefficient gcd, call it $E_{n,m}(s_3)$. In each case, division by this Gröbner basis gives remainder zero. Thus $E_{n,m}$ belongs to the rational ideal of the original equations and vanishes at every common solution. Consequently, the absence of integer roots of $E_{n,m}$ suffices to exclude the original system.

For each m the following table gives a prime modulo which $E_{n,m}$ has no root:

n	r	$\deg E_{n,m}$	rootless primes, in order $m = 0, \dots, n$
7	4	2	7, 5, 11, 7, 13, 7, 5, 7
9	6	4	11, 7, 7, 19, 11, 17, 7, 11, 7, 7
11	8	9	13, 7, 23, 13, 7, 7, 11, 29, 7, 17, 11, 7.

For instance,

$$E_{9,0}(a) = 13a^4 - 1496718a^3 + 541796355a^2 + 87999724113156a + 427746860206005240$$

has no root modulo eleven. For each of the 30 polynomials, evaluation at every residue modulo the listed prime gives a nonzero value. This excludes all three pairs. The derivations, ideal-membership checks, and residue calculations are performed by the verification program; see Section 12.

10.3. A Bernoulli-number congruence. For every case $r = n - 3$, put $\delta = n + 1 - h^0(L)$. The reciprocal Hilbert numerator has degree four, and is forced to be

$$h(z) = 1 - \delta z + (2\delta - 1)z^2 - \delta z^3 + z^4. \quad (10.4)$$

Its middle coefficient follows by evaluating $h(1) = 1$. Thus

$$B_\delta(x) = e^{2x}h(e^{-x}) = 2 \cosh(2x) - 2\delta \cosh x + 2\delta - 1.$$

In $\log \hat{A} = (n + 1) \log q + \log B_\delta$, comparison of degree $2j$ gives

$$s_{2j} = n + 1 - \frac{2j(2j)!}{B_{2j}} [x^{2j}] \log B_\delta(x). \quad (10.5)$$

We apply this identity with $2j = 12$, which is at most n for the two remaining dimensions 15 and 27.

Let $F(\delta) = 12! [x^{12}] \log B_\delta(x)$. The recurrence (4.3) gives

$$\begin{aligned} F(\delta) = & -2(39916800\delta^6 - 938044800\delta^5 + 8944523280\delta^4 \\ & - 44218331520\delta^3 + 119360516286\delta^2 \\ & - 166654367087\delta + 94009610240). \end{aligned}$$

Since $B_{12} = -691/2730$, equation (10.5) says

$$s_{12} = n + 1 + \frac{32760}{691} F(\delta).$$

The integer 32760 is coprime to 691. Integrality of s_{12} therefore forces $691 \mid F(\delta)$. Modulo 691 the polynomial factors as

$$F(\delta) \equiv -297\delta(\delta - 1)(\delta - 2)(\delta^3 + 325\delta^2 - 326\delta + 237).$$

Evaluation at the 691 residues shows that the cubic has no root in $\mathbf{F}_{691}$. Hence $\delta \equiv 0, 1, 2 \pmod{691}$. But $0 \leq \delta \leq n + 1 \leq 28$ in (10.2), so

$$\delta \in \{0, 1, 2\}. \quad (10.6)$$

The case $\delta = 0$ gives $h^0(L) = n + 1$ and is already impossible for $r = n - 3$ by Lemma 3.3. We treat $\delta = 1, 2$ by passing to a curve section.

10.4. A curve section of genus two. For $\delta = 1, 2$, we have $m = h^0(L) = n, n - 1$. Lemma 3.2 permits $n - 1$ general independent sections $s_1, \dots, s_{n-1}$ to cut a proper complete intersection curve C . Its intersection cycle has L -degree $L^n = 1$. Every irreducible component has positive integral degree, so there is a single component of multiplicity one. A complete intersection in the smooth X is Cohen–Macaulay and has no embedded associated primes. Since it is generically reduced, it is reduced everywhere: a nonzero nilpotent submodule would have an associated prime, necessarily minimal, contradicting generic reducedness. Thus C is integral and Gorenstein.

Put $A = L|_C$. Adjunction for a local complete intersection gives

$$\omega_C = (K_X + (n - 1)L)|_C = A^2, \quad \deg A = 1.$$

Therefore $\deg \omega_C = 2$ and $p_a(C) = 2$. The following argument applies to this integral Gorenstein curve even when it is singular.

To identify the section ring of the curve, let $R = \bigoplus_{k \geq 0} H^0(X, L^k)$. For every integer k and $0 < i < n$, $H^i(X, L^k) = 0$: Kodaira vanishing handles $k > -r$, and Serre duality reduces $k \leq -r$ to a nonnegative twist. The usual graded local-cohomology description of a full section ring then makes R Cohen–Macaulay of dimension $n + 1$. The $n - 1$ sections have height $n - 1$, so form a regular sequence. The quotient $\bar{R} = R/(s_1, \dots, s_{n-1})$ has dimension and depth two. The exact comparison sequence with sections on $\text{Proj } \bar{R}$ has kernel $H_{R_+}^0(\bar{R})$ and cokernel $H_{R_+}^1(\bar{R})$, both zero by that depth. Hence

$$\bar{R} = \bigoplus_{k \geq 0} H^0(C, A^k). \quad (10.7)$$

Its degree-one dimension is

$$h^0(C, A) = m - (n - 1) = \begin{cases} 1 & \delta = 1, \\ 0 & \delta = 2. \end{cases}$$

10.5. Section rings of genus-two curves. The presentations in Lemma 10.2 are those of [FPR17, Proposition 3.2]. We include a proof for integral Gorenstein curves.

Lemma 10.1. *For an integral Gorenstein curve C of arithmetic genus two, ω_C is globally generated. Its two sections define a finite flat map $f : C \rightarrow \mathbb{P}^1$ of degree two, with $f^*\mathcal{O}(1) = \omega_C$.*

Proof. Suppose a point p were a base point, and let I_p be its ideal. Then $h^0(\omega_C I_p) = 2$ and $\chi(\omega_C I_p) = 0$. Serre duality for the torsion-free sheaf $\omega_C I_p$ gives $h^0(\mathcal{H}om(I_p, \mathcal{O}_C)) = 2$.

If p is smooth, the latter sheaf is $\mathcal{O}_C(p)$. A nonconstant function with only a simple pole at p defines a degree-one morphism $C \rightarrow \mathbb{P}^1$, necessarily an isomorphism, contradicting $p_a(C) = 2$. If p is singular, put $J = \mathcal{H}om(I_p, \mathcal{O}_C)$. Locally at p it is the fractional ideal $\{g : g\mathfrak{m}_p \subset \mathcal{O}_{C,p}\}$. In fact $g\mathfrak{m}_p \subset \mathfrak{m}_p$: otherwise ga is a unit for some $a \in \mathfrak{m}_p$, and every $b \in \mathfrak{m}_p$ equals $a(gb)/(ga)$, making the maximal ideal principal and the point regular. Thus $J = \mathcal{E}nd(I_p)$, a finite sheaf of algebras. Its relative spectrum is an integral projective curve finite birational over C . Such a curve has only constant global functions, contradicting $h^0(J) = 2$. There is no base point.

The canonical map is nonconstant since $\deg \omega_C = 2 > 0$. It is finite, has degree two, and pulls $\mathcal{O}(1)$ back to ω_C . Its direct image of $\mathcal{O}_C$ is torsion-free over the smooth curve $\mathbb{P}^1$, and hence locally free. This proves flatness. $\square$

Lemma 10.2. *Let A have degree one and $A^2 \simeq \omega_C$. If $h^0(A) = 1$, then $R(C, A)$ has generators of degrees 1, 2, 5 and one relation of degree ten. If $h^0(A) = 0$, it has generators of degrees 2, 2, 3, 3 and two relations of degree six.*

Proof. First suppose $h^0(A) = 1$ and choose its section x_1 . Its zero scheme has length one and is an effective Cartier divisor. The point p is smooth: its local maximal ideal is generated by the local equation of that divisor. Thus $A = \mathcal{O}_C(p)$. Riemann–Roch and $A^2 = \omega_C$ give

$$h^0(A^0) = 1, \quad h^0(A) = 1, \quad h^0(A^2) = 2, \quad h^0(A^k) = k - 1 \quad (k \geq 3).$$

The pole orders at p of functions regular off p are therefore $0, 2, 4, 5, 6, 7, \dots$; the missing positive orders are $1, 3$. Choose functions y, z of exact pole orders $2, 5$. Every allowed pole order is $2b + 5\epsilon$ with $\epsilon = 0, 1$. Subtract a suitable multiple of $y^b z^\epsilon$ to reduce the pole order, and iterate. This proves generation by y, z for the affine function ring off p , and by the homogeneous generators x_1, y, z of degrees $1, 2, 5$ for the section ring. The square z^2 has pole order ten; reducing it gives a homogeneous relation of degree ten with nonzero y^5 coefficient. After using that relation to reduce the exponent of z to 0 or 1 , the remaining monomials in any fixed degree have distinct pole orders. They are linearly independent and form the full space counted above. Thus there are no further relations.

Now suppose $h^0(A) = 0$, and use the double cover in Lemma 10.1. The trace decomposition and Euler characteristics give

$$f_* \mathcal{O}_C = \mathcal{O} \oplus \mathcal{O}(-3), \quad f_* A = \mathcal{O}(-1) \oplus \mathcal{O}(-1).$$

For the second equality, a rank-two vector bundle on $\mathbb{P}^1$ splits as a sum of line bundles. Its Euler characteristic is $\chi(A) = 0$, so its degree is -2 ; absence of sections forces both summands to have negative degree, hence both are -1 .

Let $S = \mathbb{C}[y_1, y_2]$, with both variables of degree two, corresponding to $H^0(A^2)$. The projection formula gives, as graded S -modules,

$$R(C, A)_{\text{even}} = S \oplus Sw, \quad \deg w = 6, \quad R(C, A)_{\text{odd}} = Sz_1 \oplus Sz_2, \quad \deg z_i = 3.$$

Here $f_* A^3 = \mathcal{O}^2$ supplies z_1, z_2 . Finite-flat duality gives a perfect pairing

$$f_* A^3 \otimes f_* A^3 \longrightarrow f_* A^6 = f_* \omega_{C/\mathbb{P}^1} \longrightarrow \mathcal{O}_{\mathbb{P}^1}.$$

The equality $A^6 = \omega_{C/\mathbb{P}^1}$ follows from $\omega_C = A^2$ and $f^* \omega_{\mathbb{P}^1}^{-1} = A^4$. This pairing is symmetric. It is given by a nondegenerate constant 2×2 matrix on $\mathcal{O}^2$. Over $\mathbb{C}$ choose an isotropic basis: the self-pairings of z_1, z_2 vanish and their mutual pairing is nonzero. The trace kills the $\mathcal{O}(3)$ summand of $f_* A^6 = \mathcal{O}(3) \oplus \mathcal{O}$, so z_1^2 and z_2^2 are cubic polynomials in y_1, y_2 , while $z_1 z_2$ has nonzero w coefficient. It follows that

$$R(C, A) = \mathbb{C}[y_1, y_2, z_1, z_2] / (z_1^2 - f_3(y_1, y_2), z_2^2 - g_3(y_1, y_2)).$$

To check that the displayed relations are complete, their quotient is free over S with basis $1, z_1, z_2, z_1 z_2$. These map to a basis of the already computed free S -module $R(C, A)$. Thus the map is an isomorphism, proving the claim. $\square$

10.6. The section ring of the manifold. By (10.7), the curve generators lift to R . Together with $s_1, \dots, s_{n-1}$ they generate R : reducing a homogeneous element modulo the regular sequence and then subtracting its expression reduces the remaining degrees, so induction proves surjectivity.

Relations lift as well. For one regular element s , a polynomial relation g in the quotient evaluates in R to sb . Express b in the lifted generators and replace g by $g - sB$; this is a relation in R reducing to g . Repeat through the regular sequence. To see that these generate all relations, observe that modulo the new degree-one variables the lifted equations form the given complete intersection. Hence, together with those variables, they form a regular sequence in the polynomial ring. Its Hilbert series is the curve Hilbert series divided by $(1 - z)^{n-1}$, which is also the Hilbert series of R by regularity of the original sections. The surjection is therefore an isomorphism in each degree.

Consequently $X = \text{Proj } R$ has one of the presentations

$$X_{10} \subset \mathbb{P}(1^n, 2, 5), \quad X_{6,6} \subset \mathbb{P}(1^{n-1}, 2, 2, 3, 3). \quad (10.8)$$

The notation 1^n means n coordinates of weight one. The first ambient space has dimension $n + 1$ and one equation; the second has dimension $n + 2$ and two equations. The equality $X = \text{Proj } R$ follows for every ample line bundle by taking a sufficiently high very ample Veronese subring, which does not change Proj .

10.7. The Euler characteristic. At a point of X , consider the weights of the homogeneous generators which are nonzero there. A stabilizer of the weighted scalar action has order equal to the gcd of those weights. For all sufficiently large integers k , the line bundle L^k is generated by sections. Hence at that point, for every such k , some degree- k polynomial in the generators is nonzero. A nonzero monomial uses only generators nonzero there, so their gcd divides every sufficiently large integer. Their gcd is therefore one. Thus X lies in the free-action locus of the weighted ambient space.

On this smooth locus the weighted Euler sequence is

$$0 \longrightarrow \mathcal{O} \longrightarrow \bigoplus_i \mathcal{O}(w_i) \longrightarrow T_{\text{amb}} \longrightarrow 0.$$

The first map is the weighted radial vector field. The normal bundle of a smooth complete intersection of degrees d_j is $\bigoplus_j L^{d_j}$. Restricting the Euler sequence and using the normal sequence yields

$$c(T_X) = \frac{\prod_i (1 + w_i x)}{\prod_j (1 + d_j x)}.$$

For (10.8) this becomes

$$\frac{(1+x)^n(1+2x)(1+5x)}{1+10x}, \quad \frac{(1+x)^{n-1}(1+2x)^2(1+3x)^2}{(1+6x)^2}. \quad (10.9)$$

We compute the top coefficients by a generating-function identity. For any series $F(x)$,

$$\sum_{n \geq 0} ([x^n](1+x)^n F(x)) t^n = \frac{1}{1-t} F\left(\frac{t}{1-t}\right).$$

It follows by summing $\sum_{n \geq j} \binom{n}{j} t^n = t^j / (1-t)^{j+1}$. For the first expression in (10.9), subtracting the generating series $\sum_{n \geq 0} (n+1)t^n = (1-t)^{-2}$ gives

$$\frac{(1+t)(1+4t)}{(1-t)^2(1+9t)} - \frac{1}{(1-t)^2} = \frac{2}{5} \left(\frac{1}{1+9t} - \frac{1}{1-t} \right).$$

For the second it gives

$$\begin{aligned} \frac{(1+t)^2(1+2t)^2}{(1-t)^2(1+5t)^2} - \frac{1}{(1-t)^2} &= \frac{4}{25} + \frac{26}{75(1+5t)} \\ &\quad + \frac{4}{25(1+5t)^2} - \frac{2}{3(1-t)}. \end{aligned}$$

Extracting any positive degree n yields respectively

$$c_n - (n+1) = \frac{2}{5}((-9)^n - 1), \quad c_n - (n+1) = \frac{(12n+38)(-5)^n - 50}{75}.$$

Both are negative when $n = 15$ or 27 , contradicting $c_n = n+1$. This excludes the two remaining pairs in (10.2). This proves Theorem 1.3(i).

11. THE INDEX BOUND IN EVEN DIMENSIONS

11.1. The case $r = n - 5$. Now $c = n + 1 - r = 6$, so (10.1) gives $r \mid 90$. If n is even, r is odd. The odd divisors 1, 3, 5, 9, 15, 45 therefore give exactly

$$(n, r) = (6, 1), (8, 3), (10, 5), (14, 9), (20, 15), (50, 45).$$

Let $m = h^0(L)$ and $k = h^0(2L)$, and put

$$A = m - n - 1, \quad B = k - (n+1)m + \frac{n(n+1)}{2}.$$

Reciprocity and $h(1) = 1$ give

$$h(z) = 1 + Az + Bz^2 - (1 + 2A + 2B)z^3 + Bz^4 + Az^5 + z^6.$$

Substitution in (5.5) gives

$$\begin{aligned} N(t) &= 1 + (16A + 4B + 33)t^2 + (-16A - 8B + 27)t^4 + (4B + 3)t^6, \\ D(t) &= 1 + (64A + 16B + 137)t^2 + (-64A - 80B + 837)t^4 \\ &\quad + (-128A + 160B + 893)t^6 + (128A - 160B - 893)t^8 \\ &\quad + (64A + 80B - 837)t^{10} + (-64A - 16B - 137)t^{12} - t^{14}. \end{aligned}$$

Compute $a_{n/2}$ by (5.7); the required equation is $a_{n/2} = 1$.

For each $m = 0, \dots, n$, a prime p is specified in Appendix C. For every $k = 0, \dots, p-1$ the computed $a_{n/2}$ differs from 1 modulo p . That excludes *every* integer k , with no upper bound required. The numbers of specializations and the largest primes used are

n	r	values of m	largest prime
6	1	7	7
8	3	9	19
10	5	11	13
14	9	15	23
20	15	21	19
50	45	51	61

The 114 pairs (n, m) are therefore excluded. In particular, this proves the exclusion of $n = 10, r = 5$ used in Section 9.

11.2. The case $r = n - 7$. Here $c = 8$, and (10.1) gives $r \mid 224$. The only odd divisors of 224 are 1, 7, so the only even-dimensional pairs are $(8, 1)$ and $(14, 7)$. Section 7 already excludes the former.

For $(14, 7)$ set $m = h^0(L)$, $k = h^0(2L)$, $\ell = h^0(3L)$. The first four coefficients of $h(z)$ are

$$h_0 = 1, \quad h_1 = m - 15, \quad h_2 = k - 15m + 105, \quad h_3 = \ell - 15k + 105m - 455.$$

Set $h_4 = 1 - 2 \sum_{i=0}^3 h_i$ and reflect. The integer equation obtained from $[t^{14}]N(t)^2/D(t) = 1$, after dividing its coefficient gcd, is denoted $F_{14,7}(m, k, \ell) = 0$. It has total degree seven and 120 monomials. It is obtained by (5.5) and (5.7), and its coefficients are listed in Appendix B.

The section bound and the equality lemma give $0 \leq m \leq 14$ and $0 \leq k \leq 2^{14} + 14 = 16398$. For each pair (m, k) we test the resulting polynomial in ℓ modulo successive primes, obtaining

primes tested through	remaining pairs
initial	245985
31	4299
61	175
97	5
113	0.

Thus $(14, 7)$ is excluded as well.

Finally, if n is even and $r \geq n - 7$, parity gives just $r = n + 1, n - 1, n - 3, n - 5, n - 7$, omitting any nonpositive value. The previous section treats the first three; this section excludes the last two. This proves Theorem 1.3(ii). The proofs of C_7 through C_{10} , together with the established $n \leq 6$ theorem, prove Theorem 1.2.

12. COMPUTATIONAL VERIFICATION

12.1. The verification program. The verification program `verify_fujita_further_extensions.py` uses SymPy for polynomial algebra over $\mathbb{Q}$ and Python integers for the finite congruence calculations. It constructs the Hilbert polynomials, characteristic classes, and eliminants from the formulas in the preceding sections. The parameter bounds and moduli are specified in the source and in the tables of this article; no input data files are required.

The calculation has two parts. First, the program derives the polynomial equations using Newton's identities, Riemann–Roch, and the signature formula. It then excludes the permitted integer solutions by the discriminant and congruence tests. A successful run verifies these algebraic identities and finite calculations. The geometric arguments on which they depend are given in the preceding sections.

The principal functions correspond to the text as follows.

Calculation	Functions	Assertions checked
Dimension seven	<code>c7_symbolic_certificate</code> <code>c7_finite_certificate</code>	Chern identities, both eliminations, and discriminants.
Dimension eight	<code>c8_symbolic_certificate</code> <code>c8_arithmetic_certificate</code>	Hilbert polynomials, signature quartics, and congruences.
Dimension nine	<code>c9_eliminant</code> <code>sieve_three_parameters</code>	The resultant quartic and all bounded triples.
Dimensions ten and fourteen	<code>signature_polynomial</code> <code>compact_c10_polynomial</code> <code>check_signature_reconstruction</code> <code>sieve_three_parameters</code> <code>small_signature_sieve</code>	Signature polynomials and the corresponding congruence tests.
$r = n - 3$	<code>high_index_arithmetic</code> <code>high_bernoulli_certificate</code> <code>high_dimension_and_euler_certificates</code>	The 30 eliminants, the congruence modulo 691, and the Euler-characteristic formulas.
$r = n - 5$, n even	<code>coindex6_even</code>	The signature recurrence for all 114 specializations.

For the symbolic calculations, the program constructs the power series by (4.3) and (4.4), using rational coefficients. It solves linear equations, forms the dimension-nine resultant, and computes the Gröbner bases in Section 10.2 over $\mathbb{Q}$. Denominators and common coefficient factors are removed before an integer polynomial is passed to a congruence test.

A polynomial in one variable is evaluated modulo p by Horner's rule, at every residue $0, \dots, p - 1$, until a root is found or the residues are exhausted. A polynomial which becomes identically zero modulo p is retained, as it must be. The procedure also accommodates a drop in degree after reduction. In dimension seven, the alternative discriminant test uses the integer square root.

12.2. Representation of the candidate sets. The two largest initial sets have 102797460 and 672398100 triples. The program represents a set of third coordinates by the binary digits of a nonnegative integer. For fixed (m, k) , write

$$B_{m,k} = \sum_{\ell \text{ allowed}} 2^\ell.$$

Initially $0 \leq \ell < L$ and $B_{m,k} = 2^L - 1$. This represents all possible third coordinates for the given pair.

Fix a prime p . Whether the specialized polynomial has a root modulo p depends only on the parameter residues. For each pair $(m, k \bmod p)$, form

$$R = \sum_{\substack{0 \leq a < p \\ \text{the specialization at } \ell=a \text{ has a root modulo } p}} 2^a.$$

If $q = \lceil L/p \rceil$, the integer

$$M = R \left(1 + 2^p + \dots + 2^{(q-1)p} \right) \& (2^L - 1) = R \frac{2^{pq} - 1}{2^p - 1} \& (2^L - 1)$$

has digit ℓ equal to digit $\ell \bmod p$ of R . Here $\&$ denotes bitwise intersection. The repeated blocks have disjoint supports, so no carries occur in the multiplication. Replacing $B_{m,k}$ by $B_{m,k} \& M$ therefore applies the congruence condition to every permitted third coordinate.

The number of remaining triples is the sum of the numbers of nonzero binary digits, computed by `int.bit_count()`. Once at most 500000 triples remain, the program converts the set digits to a list

of triples and continues by direct evaluation. The length of this list is checked against the preceding count. Thus both representations describe the same finite set at every stage.

The periodic repetition formula is implemented in `repeat_residues`. The program checks it against direct enumeration for every subset of residues at moduli $1, \dots, 7$ and lengths $1, \dots, 19$. It also compares the two representations on a small range using the dimension-nine polynomial itself.

At each prime, evaluations for a repeated tuple of parameter residues are stored and reused. This is valid because evaluation of an integer polynomial commutes with reduction. For dimension nine, the substitutions (8.2) also have integer coefficients and therefore commute with reduction. The stored values are discarded when the prime changes.

12.3. Reproduction of the calculations. The source file requires Python 3.10 or later and SymPy. Running `pythonverify_fujita_further_extensions.py` derives the equations and executes all the finite checks. It prints the successive counts and ends with `ALL PYTHON CHECKS PASSED.` if every assertion succeeds. The tests use Python assertions, so the program is run without the optimization flag `-O`.

The option `-export-json audit.json` additionally writes the derived coefficient arrays, eliminants, residue values, and sieve counts. This file is an output of the calculation. The high-index eliminants have coefficient lists in descending powers. Other polynomials are recorded by their variables, monomial exponents, and integer coefficients; when present, the additional `coefficients_latex` list runs in increasing powers of the specified root variable. The program also checks its characteristic-class recurrences on $\mathbb{P}^7$, $\mathbb{P}^9$, and $\mathbb{P}^{11}$, and verifies the signature polynomials by recovering the Todd class from the Hilbert polynomial.

The repository contains the TeX source, this article, the Python file, and instructions for compiling and running them. The polynomial appendices below allow the equations and congruence tables to be read independently of the source code.

APPENDIX A. THE ELIMINANT IN DIMENSION NINE

We use the notation of Section 8: $u = s_2, v = s_4, w = s_6, a = s_3, b = s_5$. The two primitive equations obtained after eliminating s_7, s_8, s_9 are $E_i = L_i b + C_i = 0$, with the following coefficients.

$$\begin{aligned} L_1 = & -8640u^2 - 6336au \\ & + 410112u - 17280v \\ & + 103680a - 1078272 \end{aligned}$$

$$\begin{aligned} C_1 = & -14685u^5 + 324060u^4 \\ & - 31680u^3v - 240au^3 \\ & - 2326560u^3 + 590400u^2v \\ & - 34320u^2w + 2640a^2u^2 \\ & - 160320au^2 + 6453120u^2 \\ & - 9900uv^2 + 30240auv \\ & - 3127680uv + 215040uw \\ & - 53760a^2u + 890112au \\ & + 114974961216u + 162000v^2 \\ & - 362880av + 3663360v \\ & - 1920aw - 295680w \\ & + 640a^3 + 157440a^2 \\ & - 844800a - 844738332416 \end{aligned}$$

$$\begin{aligned} L_2 = & -43200u^2 + 113472au \\ & + 889344u - 86400v \\ & - 1610496a + 2350080 \end{aligned}$$

$$\begin{aligned} C_2 = & 262995u^5 - 5823780u^4 \\ & + 567360u^3v + 59280au^3 \\ & + 41549280u^3 - 8781120u^2v \\ & + 614640u^2w - 47280a^2u^2 \\ & - 559680au^2 - 112402560u^2 \\ & + 177300uv^2 - 211680auv \\ & + 39035520uv - 5859840uw \\ & + 860160a^2u - 1807104au \\ & - 2059103882688u - 702000v^2 \\ & - 362880av - 39744000v \\ & + 474240aw + 9811200w \\ & - 158080a^3 + 1109760a^2 \\ & + 678912a + 13261343095040 \end{aligned}$$

Their resultant satisfies

$$\text{Res}_b(E_1, E_2) = 241920F(u, v, w, a).$$

The coefficients of $F = \sum_{i=0}^4 A_i(u, v, w)a^i$ are

$$A_4 = 3840u - 63488$$

$$A_3 = 5760u^2 - 362496u \\ + 11520v + 2222080$$

$$A_2 = 720u^4 - 16512u^3 \\ - 4320u^2v + 152064u^2 \\ + 209664uv - 11520uw \\ + 3520512u - 2594304v \\ + 190464w - 11808768$$

$$A_1 = 13320u^5 - 242784u^4 \\ - 6048u^3v + 1837824u^3 \\ + 72576u^2v + 70272u^2w \\ - 7492608u^2 - 21600uv^2 \\ - 2953728uv - 387072uw \\ - 68163073536u + 673920v^2 \\ - 34560vw + 12146688v \\ + 141312w + 59894179840$$

$$A_0 = -12015u^7 + 765684u^6 \\ - 49950u^5v - 13992912u^5 \\ + 2029032u^4v - 28080u^4w \\ + 106964928u^4 - 59940u^3v^2 \\ - 25028928u^3v + 1415808u^3w \\ + 93693805248u^3 + 1229040u^2v^2 \\ - 56160u^2vw + 123482880u^2v \\ - 13533696u^2w - 4537368421632u^2 \\ - 16200uv^3 - 6384960uv^2 \\ + 495360uvw + 187916888448uv \\ + 41748480uw + 33647373560832u \\ + 108000v^3 + 5702400v^2 \\ - 806400vw - 1248789496320v \\ - 40857600w - 50901671137280$$

Each displayed equation is a polynomial over the integers. The resultant identity may also be verified simply by expanding $L_1C_2 - L_2C_1$.

APPENDIX B. THE SIGNATURE POLYNOMIAL IN DIMENSION FOURTEEN

For $n = 14, r = 7$, write $F_{14,7}(m, k, \ell) = \sum_{i=0}^7 B_i(m, k) \ell^i$. The variables are the section counts $m = h^0(L)$, $k = h^0(2L)$, $\ell = h^0(3L)$, not coefficients of the Hilbert numerator. The following coefficients, together with that convention, specify the entire polynomial used in the sieve.

$$B_7 = -18874368$$

$$B_6 = -7134511104m + 1453326336k \\ + 20362690560$$

$$B_5 = -1155790798848m^2 + 470877732864km \\ + 6596407590912m - 47959769088k^2 \\ - 1343814893568k - 9410722832384$$

$$B_4 = -104021171896320m^3 + 63568493936640km^2 \\ + 890365964451840m^2 - 12949137653760k^2m \\ - 362769292984320km - 2540044282281984m \\ + 879262433280k^3 + 36951535779840k^2 \\ + 517495201488896k + 2415126819309568$$

$$B_3 = -5617143282401280m^4 + 4576931563438080km^3 \\ + 64095617097400320m^3 - 1398506866606080k^2m^2 \\ - 39172524988170240km^2 - 274232905622028288m^2 \\ + 189920685588480k^3m + 7980195706306560k^2m \\ + 111741528791384064km + 521405253005328384m \\ - 9671886766080k^4 - 541906375802880k^3 \\ - 11382814465998848k^2 - 106236446111965184k \\ - 371714603757760192$$

$$B_2 = -181995442349801472m^5 + 185365728319242240km^4 \\ + 2595437832547860480m^4 - 75519370796728320k^2m^3 \\ - 2114962182037831680km^3 - 14803616548748132352m^3 \\ + 15383575532666880k^3m^2 + 646287634417582080k^2m^2 \\ + 9048032227521921024km^2 + 42212600683249065984m^2 \\ - 1566845656104960k^4m - 87774136636538880k^3m \\ - 1843398338061828096k^2m - 17201655546463420416km \\ - 60177407870074533248m + 63834452656128k^5 \\ + 4470319371386880k^4 + 125188081996054528k^3 \\ + 1752420756180602880k^2 + 12262096027360696000k \\ + 34310884872797383402$$

$$\begin{aligned}
B_1 = & -3275917962296426496m^6 + 4003899731695632384km^5 \\
& + 56052068529261772800m^5 - 2039023011511664640k^2m^4 \\
& - 57094416397290700800km^4 - 399563744123097317376m^4 \\
& + 553808719176007680k^3m^3 + 23262458998475980800k^2m^3 \\
& + 325620044334329757696km^3 + 1518889838325936979968m^3 \\
& - 84609665429667840k^4m^2 - 4739009781222604800k^3m^2 \\
& - 99510166256361209856k^2m^2 - 928422555999352602624km^2 \\
& - 3247402009645454876928m^2 + 6894120886861824k^5m \\
& + 482713662770380800k^4m + 13515784244957085696k^3m \\
& + 189166326551280304128k^2m + 1323418788249959729408km \\
& + 3702470602277503658824m - 234059659739136k^6 \\
& - 19667609026560000k^5 - 688408640537829376k^4 \\
& - 12847561883227873280k^3 - 134833727662115152448k^2 \\
& - 754494298636214961516k - 1758664553769502206697 \\
\\
B_0 = & -25271367137715290112m^7 + 36035097585260691456km^6 \\
& + 504384118879407833088m^6 - 22021448524325978112k^2m^5 \\
& - 616516421899247419392km^5 - 4313842577830193135616m^5 \\
& + 7476417708876103680k^3m^4 + 313990602631906590720k^2m^4 \\
& + 4394397830215019397120km^4 + 20494705844786211127296m^4 \\
& - 1522973977734021120k^4m^3 - 85287891313297981440k^3m^3 \\
& - 1790582921129257205760k^2m^3 - 16703206200200828583936km^3 \\
& - 58414055525899661578752m^3 + 186141263945269248k^5m^2 \\
& + 13031086502636421120k^4m^2 + 364803926672740515840k^3m^2 \\
& + 5104923740232864915456k^2m^2 + 35708356992589914385152km^2 \\
& + 99882939615801221882664m^2 - 12639221625913344k^6m \\
& - 1061873062887555072k^5m - 37161614273488404480k^4m \\
& - 693419662686368464896k^3m - 7276141194137848482176k^2m \\
& - 40708552503242224866488km - 94872385352327729726730m \\
& + 367808036732928k^7 + 36053990168199168k^6 \\
& + 1514222256517341184k^5 + 35321107231167932416k^4 \\
& + 494209522557826902080k^3 + 4147821048038964926314k^2 \\
& + 19334692281325631968759k + 38615228954566746978894
\end{aligned}$$

APPENDIX C. CONGRUENCE MODULI FOR $r = n - 5$

Each row says: for the indicated n and each listed value of m , the polynomial $a_{n/2} - 1$ from the recurrence in Section 11.1 has no root k modulo the indicated prime. The lists for each n partition $m = 0, \dots, n$.

n	Prime	Values of m
6	3	1, 2, 4, 5
6	5	0, 3
6	7	6
8	7	2, 3, 5, 6
8	11	0, 1, 4, 8
8	19	7
10	3	0, 1, 3, 4, 6, 7, 9, 10
10	7	2, 5
10	13	8
14	5	1, 2, 6, 7, 11, 12
14	7	4, 5, 8
14	11	0, 3, 14
14	23	9, 10, 13
20	3	1, 4, 7, 10, 13, 16, 19
20	7	3, 17
20	11	2, 12, 15
20	13	5, 6, 18
20	17	8, 9, 11, 14, 20
20	19	0
50	5	1, 6, 11, 16, 21, 26, 31, 36, 41, 46
50	7	4, 5, 8, 12, 15, 18, 19, 22, 25, 29, 32, 33, 39, 40, 43, 47, 50
50	11	0, 2, 10, 13, 24, 27, 35, 38, 44, 49
50	13	34
50	17	7, 28, 30, 45
50	19	37, 48
50	23	14, 23
50	29	20
50	37	42
50	47	3, 17
50	61	9

APPENDIX D. SIEVE TABLES

The prime in a row is applied after all preceding primes in that column. A dash means that prime was not used in that particular sieve. The initial counts are 102797460 for C_9 and 672398100 for C_{10} .

Prime	Remaining triples for C_9	Remaining triples for C_{10}
3	—	448265745
5	—	267329529
7	—	181171361
11	—	119711405
13	71051845	80209438
17	47418292	51982239
19	31752841	33510036
23	20583262	21301530
29	13643896	13548080
31	8899360	8661101
37	5721568	5489690
41	3671815	3474806
43	2357661	2211620
47	1511609	1397055
53	964275	887441
59	611168	568511
61	390389	360233
67	248724	227836
71	157634	144692
73	99718	91881
79	63012	58870
83	40028	37415
89	25330	23681
97	16021	14926
101	10047	9501
103	6324	6032
107	3958	3869
109	2444	2403
113	1501	1511
127	938	946
131	579	614
137	357	409
139	217	254
149	136	148
151	87	90
157	57	63
163	39	41
167	25	28
173	14	18
179	10	12
181	3	7
191	3	3
193	2	3
197	2	2
199	1	1

Prime	Remaining triples for C_9	Remaining triples for C_{10}
211	1	1
223	0	1
227	—	1
229	—	1
233	—	1
239	—	0

For $n = 14, r = 7$, the complete pair counts are:

Prime	Remaining pairs
2	245985
3	191325
5	160717
7	101680
11	63341
13	41844
17	28247
19	18531
23	10837
29	6812
31	4299
37	2733
41	1681
43	1069
47	659
53	415
59	249
61	175
67	117
71	68
73	40
79	24
83	13
89	10
97	5
101	4
103	2
107	2
109	1
113	0

APPENDIX E. AN ELIMINANT MODULO ELEVEN

The $n = 9, r = 6, m = 0$ eliminant displayed in Section 10.2 has the following values modulo eleven. All entries are nonzero, so this specialization has no integer root.

a	0	1	2	3	4	5	6	7	8	9	10
$E_{9,0}(a) \bmod 11$	10	8	7	6	8	9	9	1	4	8	7

The optional JSON export records every coefficient vector and every residue list for all 30 high-index eliminants. In each coefficient vector the coefficients are ordered from the highest power down to the constant term. The accompanying program derives these polynomials from the Hilbert polynomial.

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